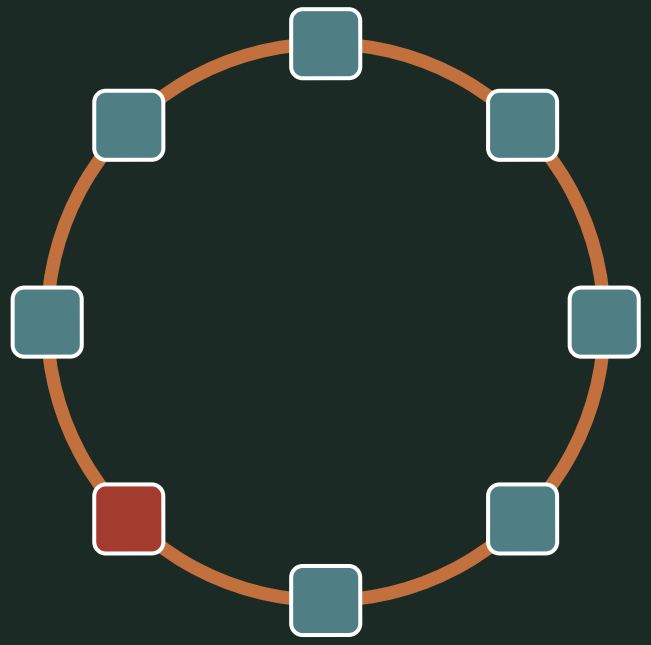
The Chip That Says No

A multi-layer hardware approach to AI safety. Each topic in two slides: the plain version with a picture, then the technical one.



The halt loop: one series current loop through every governor in a rack. Any governor opens it, or a cut wire, and every GPU halts.

Yobie Benjamin · yobie AT ieee DOT org · linkedin.com/in/yobie

What this project is

A design for a small safety chip that sits beside every AI GPU and holds the only stamp a device needs before it acts. Built in the open as a model with tests, hardware blocks and a circuit simulation. No silicon yet, by choice.



done: every block as software, every attack as a test

in progress: invariants written, proofs next

written: documents 00 to 18, RTL blocks, loop circuit

after 3, on a real PCIe or Ethernet path

Two open repositories

hardware-and-silicon: the software reference stack and the blog series. safety-governor: the chip design, model, tests, RTL, silicon flow.

One rule for the program

Nothing goes to hardware until the design is proven in software, firmware and circuit simulation: 1000 percent sure in the pre-hardware phase, because tape-out is where mistakes get expensive.

Plan of record: the rack tier

The chip rides on the server maker's management card and reaches NVIDIA's GPU board through its connector. Nothing changes in the GPU or in CUDA.

THE PICTURE
the multi-layer post, roadmap

The same catalogue of attacks (75 today, with 12 legitimate baselines) is the acceptance test for every phase, from the software reference to the FPGA. The catalogue is the constant; the artifact changes.

Thesis

Safety that lives only in software can be removed by software.

So the decision to act has to be made in silicon the model cannot reach, at the few places where computation turns into effect.

Every layer above the chip is a witness. Only the layer built into silicon can be the lock.



Witness

can keep a gate shut

- Safety training
- A judge model
- A watermark
- An observer
- A log read later



Lock

is the gate

- A ticket the device checks for itself
- A switch port that stays shut
- A response held at the exit
- A breaker that opens

THE PICTURE
the multi-layer post

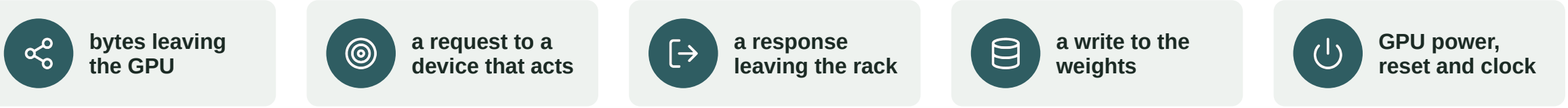
A judge model is a witness. It can keep a gate shut. It can never be the gate. The layers above the chip are witnesses, and the only layer that can be the lock is the one built into silicon.

The thesis, technically: a reference monitor

THE THREE TESTS OF A REFERENCE MONITOR: TAMPERPROOF, ALWAYS INVOKED, SMALL ENOUGH TO VERIFY

Property	What it means here	How the governor meets it
Tamperproof	The host, the model and the operator cannot alter the decision logic or the keys	A separate die. Device secret in fuses with no read port. Boot from ROM, extend-only measurement registers, an anti-rollback counter. The rulebook changes only with k-of-n signatures from keys held outside the rack.
Always invoked	There is no path around it	A comparator at every crossing: switch ports open only for roster peers; a device acts only on a ticket; the egress holds a response until a ticket releases it; weights are written only with a ticket; power, reset and clock answer to the governor. Default deny; fail closed.
Verifiable	Small enough that its correctness can be checked	About 2 mm² of core logic as configured (1.4 by 1.4 mm). Invariants written down: no ticket without a rule and its witnesses, no bytes to a stranger, halt on silence, no governor clears its own halt. 200 model tests, 20 RTL tests, an independent audit.

THE FIVE CROSSINGS THE CHIP STANDS AT



THE PICTURE
docs 00 and 18

A firewall for actions, with a hand on the breaker: it decides what leaves, and when it must, it also pulls the plug. A chip that can only ask the GPU nicely is a classifier with a heatsink.

The logic behind it

The customs officer

She cannot read the language on the paperwork. She does not need to.



the seal — unbroken?

the manifest — matches the count?

the scale — matches the weight?

the list — destination approved?

Smuggling is caught as a mismatch between what the papers say and what the scale says, not by understanding the cargo. The governor works the same way: on structure, never on meaning.

Software safety can be edited out

Refusal lives mostly along one direction in the activation space. Subtract it from the weights and the model answers everything. It takes an afternoon on a consumer GPU.

Watchers can only warn

A judge model is a transformer too: it falls for the same injections, can be edited the same way, and runs as a process the operator can patch or kill.

Hardware checks structure, not meaning

A ticket, a signature from a key the model does not hold, a count under a limit, a peer on a list, a measured workload. A chip can refuse on those.

THE PICTURE
the multi-layer post; docs 17

So: put the decision in silicon, and make the rulebook an allow-list, signed, default deny. No rule, no path. There is no rule for exporting weights.

The logic, technically: what a chip can see

Seen at a crossing (small, exact, checkable)

- The attested workload: a hash of GPU firmware, host firmware and model weights, measured out of band
- The action and its typed fields: strike with a target and a munition; payments.transfer with an amount and a destination
- Counts of tokens, cycles and bytes from the GPU's own counters
- Whether the peer behind a switch port is on the roster
- The rules edition every message was signed under
- Signatures: an observer's verdict over a hash, a human's approval over a hold
- What the switches actually carried, to cross-check what was presented

Never seen

The meaning of a token. Intent. The chain of thought. Nothing in the rack can look at a stream of integers and say "that is a weapon". A design that promises to catch bad words on the wire promises what no hardware can do.

So meaning is delegated, with three guarantees

- To signed lists (protected sites, approved recipients) and to observers (classifiers, people with keys) that sign verdicts over hashes
- Always invoked: no verdict, no ticket. Cannot be impersonated: a verdict under the host's key, or the governor's own, is refused. Separate: a different key, a different box, outside the model's reach

THE PICTURE
docs 17, section 1

Two bridges. Tokens and arithmetic bridge the software and the GPU. Hashes, signatures, counters, identities and typed fields bridge the model and the governor. The second bridge is the only one the chip walks on.

The proposal: a referee chip beside every GPU

The model, on NVIDIA GPUs

Computes freely. Nothing it thinks has any effect until bytes leave or a request is made.

request →

The governor

Checks who is asking, whether a rule allows it, whether the witnesses agree, whether the budget holds. Then signs a ticket, or says no.

ticket →

The things that act

A weapon, a payment switch, a message gateway, the storage that holds the weights, the rack's own power. Each checks the ticket itself.



Added, not redesigned

It rides on the server maker's management card. Nothing changes in the GPU or in CUDA.



An allow-list, not a blacklist

Signed by a k-of-n quorum whose keys sit outside the chip vendor and the operator. Everything not on it is no.



A hand on the breaker

One series loop through every governor in the rack. Any one opens it, or a cut wire, and every GPU halts.



Buildable now

At the rack tier, with a server maker. The tighter tiers are asks to NVIDIA, firmware and board work, not prerequisites.

THE PICTURE
docs 18, section 1

A referee with a whistle and a rulebook he did not write: the chip holds the whistle, the rulebook is a signed document handed to it. What is burned in is how to enforce; what is loaded is what is allowed.

Proposal, technically: the module and the die

Where it sits and what it reaches

- One governor per GPU slot on the OEM's management-module card (DC-SCM class)
- Per slot through the HGX connector: SMBus (SMBPBI), power brake, PERST#, PWR_EN, present and fault. HGX reset is grouped today (three PERST inputs for eight GPUs)
- NVSwitch port disable out of band through NSM; DPU steering tables through DOCA on the management port
- The GPU's confidential-computing attestation taken as an input; the BMC reads through the governor

What is on the die

- An OpenTitan-class root of trust: fuses, DICE key chain, OTBN for P-384 signatures (1,697,985 cycles, 17 ms at 100 MHz, so one signature per 50 ms batch), AES-256-CTR with HMAC-SHA-256 for storage
- A mailbox of 21 signed commands, checked in order: sequence number, rules edition, signature; the first failure is reported
- The policy image and its epoch; the gate core (lookup and comparators); the ledger (budgets, rate windows); a hash-chained log whose head lives in a register
- Meter, heartbeat, lever drivers, loop I/O and the signed message fabric between governors

Tier	Where the governor's authority comes from	Status
Rack	Management-card levers, switch and DPU management paths, the halt loop	Plan of record; needs a server OEM, nothing from NVIDIA's silicon
Board	Same firmware, policy format and fabric on a board without a baseboard connector of its own	Fallback; the gaps it leaves are recorded, not hidden
Integrated	Buffer-class checkers in the GPU's copy engines and peer MMU; per-GPU reset	Needs NVIDIA; closes the one gap the rack tier records (H2b)

The layers: witnesses above, the lock below

5	**Governance root**	Signs the rulebook, the approved-firmware list and revocations. Separate from the chip vendor on purpose	LOCK
4	**Control plane: the gate**	Default deny on the only path out: attestation, rulebook, witnesses, budget, human holds, one-time tickets, tamper-evident audit	LOCK
3	**Observers**	Fast, cheap early-warning verdicts about one request, signed, consumed as witnesses	WITNESS
2	**Provenance**	Watermark, hardened detector, retrieval, signed registry; emits a signed provenance verdict	WITNESS
1	**Model and runtime**	The model, CUDA, the serving stack. Treated as untrusted; nothing here is allowed to decide anything	UNTRUSTED
0	**Silicon root of trust**	Fused device secret, anti-rollback counter, extend-only measurement registers, DICE key derivation, quote engine	LOCK

THE PICTURE
the multi-layer post, layer 3

Biology separates the alarm from the muscle. The spinal reflex pulls the hand off the stove before the brain knows; the early warning is fast, cheap and often wrong, and the system tolerates that because the warning never acts on its own. The layers above the chip are reflexes. The chip is the muscle.

Layers, technically: mechanism, platform

Layer	Mechanism	What it signs or checks	On NVIDIA's platform
5 Governance root	k-of-n quorum over the policy image; rollback refused; expiry enforced	The rulebook (allow-list), the approved-firmware list, revocations	The policy authority's keys; OpenShell's policy prover checks the derived network rules
4 Control plane	Attestation check, rule lookup, witness collection, budget and rate, human hold, one-time ticket bound to a hash of the fields, hash-chained log	Tickets, log checkpoints, the compliance report	OpenShell supervisor middleware today in software; the governor module in hardware; a DOCA application on BlueField-4 next; an I/O-stage block in the GPU eventually
3 Observers	Verdict over the hash of one request: risk, flags, issued and expiry times, under an anchored key	A signed verdict the gate requires on consequential actions	Sentry on BlueField-4 as the required witness; an independent observer as the optional second
2 Provenance	Watermark, hardened detector, retrieval, signed registry	A verdict {registry, content digest, exact | tampered | unknown}	Runs beside the vendor's sampler; its verdict is a witness the gate can require
1 Model and runtime	Measured into the workload hash; routed through the switches and egress the governor controls	Nothing. It can sign nothing the gate accepts	The GPU, CUDA and the serving stack, unchanged
0 Silicon root of trust	Fused device secret, anti-rollback counter, extend-only measurement registers, DICE derivation, quote with a fresh nonce	Quotes: the measurement, the key chain, the counter	The GPU's confidential-computing attestation and the DPU's own root of trust, bound together

How it meets the model

1 The model thinks

Token ids in, numbers out. No authority here and none needed.

2 The runtime asks

It turns the output into a typed request: the action and its fields.

3 The governor checks four things

Who is asking. Is there a rule. Do the witnesses agree. Is the budget intact.

4 Ticket, hold, or no

A one-time signed ticket bound to a hash of the exact fields. Or a hold for a human. Or a refusal, logged.

5 The device checks the ticket

Against what it measures itself. A changed amount or target breaks the hash.

6 Act, and record

The action happens and the record goes into a hash chain the chip signs.

THE PICTURE
the multi-layer post, layers 0, 4 and 5

The endpoint that moves the money verifies the ticket itself. Change the amount after the ticket was issued and the hash breaks. Present it twice and the second is refused. The model is never the one who decides.

The flow, technically: every check, every refusal

Step		What is checked	Fails with (as the model prints it)	Backed by
1	**Mailbox**	Sequence number, then rules edition, then signature; the first failure is reported	mailbox.bad_sequence, mailbox.bad_signature; a frame from an old edition is refused before the gate sees it	H3: test_mailbox_reports_the_first_failing_check
2	**Who is asking**	The workload hash (GPU firmware, host firmware, model) and its evaluation attestation	model.not_evaluated	H1 attestation; H18 oversight
3	**Is there a rule**	Exact rule beats wildcard; typed limits; signed lists for targets and recipients	gate.constraint.target.in_denied_list, endpoint.target_protected	H6 comparator; the strike narration
4	**Do the witnesses agree**	Observer verdict from a key in the observer set, fresh, under the risk limit; human quorum on holds	gate.witness_missing, gate.held_for_human, release.hazard_class_not_allowed	H17 use cases; H18 oversight; H8 override
5	**Is the budget intact**	Compute budget and license, session hazard budget, byte budgets, rate windows	release.session_hazard_budget_exceeded, compute.budget_exceeded	H13 shared budget; H18
6	**The ticket**	Signed over the hash of the exact fields; one use; short expiry; bound to one endpoint	ticket.missing, ticket.replay, ticket.bad_signature, ticket.wrong_endpoint	H17: forged, replayed and wrong-hash tickets never deliver
7	**The crossings**	Switch ports open only for roster peers; the egress holds responses until released; bytes cross-checked against what was presented	ep.no_capability, gate.output_unpresented	H2 enforcement points; the plot narration

Where it stands: drawings tested, no silicon yet

200

hypothesis tests pass in the functional model of the governor and a simulated 72-GPU rack (H0 to H18)

20

simulation tests pass on the hard-core RTL blocks, with a LibreLane flow to the open 130 nm process

317 µs

worst case for the halt loop to reach the last of 72 governors, against a 1 ms bound, across 39 simulated configurations

~2 mm²

of core logic for the governor as configured, 1.4 by 1.4 mm, inside a 10 mm² shuttle harness

75

attacks blocked with the predicted reason code in the software reference stack, with all 12 legitimate baselines passing (results/report.md)

0

changes needed in the GPU, in CUDA or on the baseboard at the rack tier

THE PICTURE
docs 18, section 3

Everything here is model, RTL simulation and circuit simulation. The two numbers that decide adoption, the real overhead and the false-halt rate, are not measured on hardware yet.

The challenges, stated plainly



Reach: the governor's authority ends at the last device that verifies its signature. A payment system is two or three hops away and has never heard of it.

Meaning

The chip guarantees an observer is always invoked. It cannot make the observer right about a disguised request.

False halts

The first support ticket will read: the governor halted my rack. The rate is not measured yet.

The levers

On today's HGX board, reset is grouped and the per-module signals sit in a design guide under NDA.

Cost

One P-384 signature costs 17 ms on the OTBN engine. Batching keeps it under one percent in the model, not yet in silicon.

Governance

Who holds the quorum keys, whose policy governs a shared rack, and the integrated tier needs NVIDIA.

Adoption

A chip wired to nothing that acts is a logger. The first device maker to check a stamp decides this.

THE PICTURE
docs 18, section 5

The perimeter firewall decides whether the packet leaves; it does not decide what the server on the far side does with it. The answer is to move the credential behind a ticket-checking adapter, not to reach through the bank.

Challenges, technically: mechanism, status

Challenge	What the design does	Status
Reach: a payment system two or three hops out	A: the comparator in the device. B: move the credential behind a ticket-checking adapter, so the model's runtime never holds the API key. C: open the DPU flow to the provider only under a live ticket. D: always the signed record	A and C modelled and tested (H6, H2, H17). B is a pattern the endpoint model already expresses; the three-hop payment narration is the next piece of work
Meaning: observers can be fooled	Observer always invoked, cannot be impersonated, separate from the model; hazard classes, legibility floor, session hazard budget, TRNG audit sampling with escrow	Guarantees tested (H17, H18). Observer error rates: open; needs a red team and a quorum of observers on high-hazard classes
False halts	Loop beacon inside the signed heartbeat; loss of current declared after two sense periods (debounce); a jumpered governor caught within two heartbeats; FMEA	Modelled (H11) and simulated (ngspice). Not measured on hardware
Levers on HGX	Per-slot SMBus, power brake, PERST#, PWR_EN at the connector; grouped reset accepted as coarser; the evidence register names what is verified	Evidence register done. Design guide under NDA still needed
Cost	P-384 on OTBN at 1,697,985 cycles (17 ms); one signature per 50 ms window; engine 34 percent busy batched, 272 percent unbatched; firmware share under 0.1 percent of GPU time	Modelled (H9). Silicon or FPGA measurement: open
Governance and adoption	k-of-n quorum with keys outside vendor and operator (goal G8); partitions and per-rule approvers; integrated-tier asks written	Two tenants with two quorums on one baseboard: not designed. Integrated tier: needs NVIDIA

What I need to prove or disprove the thesis

A server maker to talk to

About the management-card slot and the HGX design guide.

A device maker willing to check a ticket

A functional-safety vendor, or whoever runs the tools an agent calls.

A lab with a rack and an FPGA

To measure the real overhead and the false-halt rate over weeks.

A red team

To attack the release gate and the observers with disguised requests.

Chip designers and an auditor

To tear the design apart before any tape-out.

THE PICTURE
the program's own rule

Going to hardware and then tape-out is so expensive that we need to be 1000 percent sure in the pre-hardware phase. These five are what proving, or disproving, the design takes before that. A no is informative.

What I need, technically: who, what, and why

Who	What I need	What it settles
A server OEM	The management-module slot; the HGX design guide under NDA; the connector pinout	Whether per-module levers reach the connector. If not, the rack tier is coarser than the design says
NVIDIA	Seven firmware asks: NSM port-disable semantics, per-GPU reset, buffer-class checkers in the copy engines and peer MMU	The integrated tier; whether the recorded gap (H2b) can be closed from inside the silicon
An endpoint partner	A device or tool gateway that verifies the ticket: P-384 signature, typed fields, epoch, freshness	Whether the comparator works outside the model. If nobody will check the stamp, the thesis fails in practice
A lab with an HGX or NVL rack	An FPGA governor beside a real GPU for weeks	The real overhead against the one percent goal (G6); the false-halt rate; the loop beacon in the field
A model lab's red team	Disguised requests against the release gate and the observers	The size of the semantic residual; whether a quorum of observers is enough
Chip designers, an auditor	Review of the RTL, the firmware plan and the proof-assumption list; formal proofs of the block invariants	Verification before tape-out

What would disprove it

- No device that acts will verify a ticket
- The levers do not reach the baseboard connector
- False halts at a rate operators will not accept
- Measured overhead above one percent after batching
- Observers miss disguised requests too often for the hazard cases
- Nobody will hold the quorum keys

In one sentence

A separate safety chip beside every AI GPU that checks each action the AI attempts against rules signed by people the AI cannot influence, stamps the allowed ones so the devices that carry them out can verify the stamp themselves, and cuts the power when something is wrong.

The AI can think whatever it likes. It cannot act without the stamp, and it cannot forge one.
Added to the server's own boards. Nothing changes in the GPU or in CUDA.

github.com/YobieBenjamin/safety-governor · github.com/YobieBenjamin/hardware-and-silicon
The Chip That Says No, parts 1 to 4, and design documents 00 to 18

Yobie Benjamin · yobie AT ieee DOT org · linkedin.com/in/yobie